Project 2 - Privacy vs Utility

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Introduction

Chosen data set:
https://www.kaggle.com/datasets/osmi/mental-health-in-tech-survey/data

This dataset contains responses from a 2014 international survey conducted by Open Sourcing Mental Health regarding mental health experiences in the workplace for those who work in tech. We identified the following columns as quasi-identifiers: ***Age, Gender, Country, State*** *if Residing in US*, and ***Number of Employees*** *at Workplace*. These attributes are considered quasi-identifiers as they do **not** directly identify an individual but in combination with other data sources, these attributes could lead to the re-identification of a respondent. Additionally, these quasi-identifiers could be found through some other data source by an adversary. Although there were no identifiers in this publicly released data, the remaining survey questions pertained to more personal matters such as family history of mental health or if respondents had sought help for mental health. We regarded all such columns as sensitive information as we assumed that this information would not be able to be found through some other data source by an attacker.

Among the original columns containing sensitive information, we selected **work_interfere** ("If you have a mental health condition, do you feel that it interferes with your work?") as the primary sensitive attribute. A sensitive attribute is one that contains private information about individuals that should be protected and is not easily inferable from other public sources by an adversary. We focused on this column because it is both categorical and diverse, meaning that enforcing ℓ-diversity on this attribute requires meaningful variation across equivalence classes. All other sensitive attributes were dropped for

the purpose of this analysis, to concentrate the privacy evaluation and reduce complexity when computing utility metrics.

Data Tidying

As this dataset contained actual survey responses, the data required some cleaning before it was able to be anonymized. Several fields such as ***Age, Country***, and ***Gender*** were designed as open-ended responses, resulting in rather messy data including nonsensical answers *(Age = 9999999)* and typographical errors *(femail rather than female)*. We recognized gender as an open-ended field so we considered all responses as legitimate responses. With unique open-ended responses for this field, it was necessary to generalize for usability. We defined usability as the ability for the data to be utilized in statistical analysis. We categorized responses into *Female, Male*, and *Other*. We considered this as part of the data-cleaning process. Although we did generalize the attribute, we did not include this loss of granularity as part of our utility metric as it was necessary for the dataset to be usable.

With other fields such as ***Age, Country***, and ***State***, we decided that we needed a legitimate response for both fields for a record to be considered to be complete. If these two fields were left incomplete or had a nonsensical response, we discarded that row. For age, we considered ages from ***17 - 100*** as legitimate. After removing incomplete records, the dataset went from containing ***1260*** records to ***1251*** records. This tidied version of the raw data is saved as *survey_data_clean.csv*. In this project, we achieved k-anonymity for this dataset for $k = 2, 3, 4$, along with higher values of k, when we begin our work in ℓ-diversity.

Initial K-Anonymization

Overview of K-Anonymity

K-anonymity is a privacy technique that ensures each record in a dataset is indistinguishable from at least $k - 1$ others with respect to a set of quasi-identifiers (**QIDs**). The goal is to prevent re-identification by grouping records

into equivalence classes of size k or greater. In this project, we applied k-anonymization to protect individuals' privacy while preserving useful information. We began by anonymizing the dataset to $k = 2, 3, 4$.

K = 2

With the QID set $\{\mathbf{Age}, \mathbf{Gender}, \mathbf{Country}, \mathbf{State}, \mathbf{NumEmployees}\}$, we first checked whether the dataset satisfied 2-anonymity using the pandas `duplicated` function. Many records failed this test, so we dropped the **State** column to reduce uniqueness—people living in specific U.S. states could potentially be identified when combined with **NumEmployees**.

After removing **State**, we treated **NumEmployees** as a sensitive attribute instead of a QID, as it remained important for understanding workplace mental health. To increase equivalence class sizes, we generalized key attributes:

- **Age**: Binned into $(17, 26), (26, 30), (30, 50), (50, 100)$.
- **Country**: Grouped all countries outside the U.S., Canada, and U.K. into regions (*Europe* and *Asia/Africa/Americas*).
- **Gender**: Combined *Female* and *Other* into *Female/Other* due to small sample sizes.

These transformations produced a dataset satisfying 2-anonymity.

K = 3

Starting from the 2-anonymous dataset, we found the following equivalence classes still had only two members:

- Female/Other, 30–50, Africa/Asia/Americas
- Female/Other, 50–100, United States

To reach 3-anonymity, we adjusted age ranges to $(17, 26), (27, 44), (45, 100)$, which successfully increased group sizes to at least three.

K = 4

Testing the 3-anonymous dataset revealed one remaining equivalence class of size three:

- Male, 44–100, Europe

To achieve 4-anonymity, we grouped the United Kingdom into the *Europe* region, merging records and ensuring all equivalence classes contained at least four members.

Utility Metric: KL Divergence

To evaluate the information loss from k-anonymization, we used *KL divergence* as a utility metric. *KL divergence* measures the distance between two empirical distributions:

- $P(x)$ — the empirical probability of each quasi-identifier (**QID**) combination x in the original dataset D, estimated by $F'(x)$ as the fraction of tuples matching x;
- $P_{\text{anon}}(x)$ — the probability of x in the anonymized dataset D', estimated by
$$P_{\text{anon}}(x) \;=\; \frac{1}{N}\sum_{C}\mathbf{P}[x \in C]\,|C|$$
where the sum is over all equivalence classes C, $|C|$ is the size of class C, and N is the total number of records.

The KL divergence between the original and anonymized distributions is then
$$D_{KL}\big(P \,\|\, P_{\text{anon}}\big) \;=\; \sum_{x\in\mathcal{X}} P(x)\,\log\frac{P(x)}{P_{\text{anon}}(x)}$$
where $\mathcal{X}$ is the set of all QID combinations in the dataset.

In practice, some quasi-identifier combinations may appear in the original dataset but not in the anonymized dataset, which would make $P_{\text{anon}}(x) = 0$ and cause the logarithm in D_{KL} to be undefined. To avoid infinite or unstable values and to keep the results interpretable, we applied a small smoothing constant ε (epsilon) when estimating probabilities:
$$P(x) \;\leftarrow\; \frac{F'(x)+\varepsilon}{N+\varepsilon|\mathcal{X}|}, \quad P_{\text{anon}}(x) \;\leftarrow\; \frac{F'_{\text{anon}}(x)+\varepsilon}{N+\varepsilon|\mathcal{X}|}.$$
This smoothing ensures that no probability is exactly zero and produces KL divergence values that remain finite and comparable across different k settings,

to make the results easier to comprehend.

Below are our results using KL divergence as a metric of Utility for joint QIDs at the different levels of our intial anonymization at the different values we chose for k:

k	**Joint KL Divergence**
2	24.417195
3	24.750623
4	24.750623

Table 1: Joint KL divergence results for $k = 2, 3, 4$ -anonymization across Age Range, Gender, and Country.

ℓ-diversity

Values of ℓ at 2, 2.5, and 3

The ℓ-diversity principle is a privacy measure that ensures each equivalence class contains at least ℓ "well-represented" values for a sensitive attribute. A table is ℓ - diverse if every equivalence class is ℓ - diverse. This improves privacy protection guarantees for our anonymized dataset as it address the privacy concern of sensitive value uniformity that is not addressed by k-anonymity.

We utilize the Entropy ℓ-diversity that is defined as follows:
A table is Entropy ℓ-diverse if for every equivalence class c

$$\sum_{s \in S} p_{(c,s)} \log(p_{(c,s)}) \geq \log \ell$$

- $s \in S$ - the set of sensitive values for a sensitive attribute S
- $p_{(c,s)}$ - the probability of the sensitive value in the equivalence class or the fraction of tuples in the equivalence class with sensitive attribute value equal to s

$$p_{(c,s)} = \frac{n_{(c,s)}}{\sum\limits_{s' \in S} n_{(c,s')}}$$

We selected ℓ values of 2, 2.5, and 3 to explore increasing levels of diversity in the sensitive attribute, allowing us to evaluate how stricter privacy constraints

impact utility while keeping equivalence classes reasonably sized.

In order for us to match our values of ℓ-diversity, we employed further generalization and suppression of quasi-identifiers to ensure that sensitive value distribution within an equivalence classes was ℓ diverse at the values 2, 2.5 and 3. Due further anonymization, the values of our initial $k = 2, 3, 4$ changed to $k = 19, 30, 85$ to achieve ℓ-diversity. We chose these k values by generalizing until ℓ diversity was met for $\ell = 2, 2.5, 3$ for each dataset and then set the minimum group size among the equivalence classes as our k value. The following transformations were made to ensure ℓ-diversity for each equivalence class for every ℓ value and every k value.

- **Country**:
  - $k = 19$: Suppressed Country Attribute by inserting value "Country" for all entries
  - $k = 30, 85$: Generalized Country into "North/South America" and "Eurasia-Africa-Oceania"
- **Gender**:
  - $k = 19$: Combine Gender categories ("Female" + "Other") resulting in categories *"Female/Other"* and *"Male"*
  - $k = 30, 85$: Combined all genders into one "*Male/Female/Other*"
- **Age**:
  - $k = 19$: Generalize Age into ranges $(17-28), (28-33), (34-40), (41+)$
  - $k = 30$: Generalize Age into ranges $(17 - 26), (27 - 40), (41+)$
  - $k = 85$: Generalize Age into ranges $(17 - 35), (36+)$

ℓ- Diversity Values

The Entropy ℓ thresholds or $\log \ell$ for each ℓ are displayed below:

	$\log_2 \ell$
$\ell = 2$	1.0
$\ell = 2.5$	1.32
$\ell = 3.0$	1.58

Table 2:

We calculated the minimum and average Entropy ℓ values for each value of k to ensure that each dataset met the above thresholds for $\ell = 2, 2.5, 3$. The results are displayed below:

	$k = 19$	$k = 30$	$k = 85$
min ℓ	1.6409	1.6802	1.7122
mean ℓ	1.7957	1.8065	1.8081

Table 3: Minimum and Mean Entropy ℓ diversity values for $k = 19, 30, 85$

KL- Divergence Values

Following the increase in k values ($k = 19, 30, 85$) due to meeting ℓ-diversity, we see that our KL divergence results are relatively similar to prior KL divergence values for $k = 2, 3, 4$:

k	ℓ **= 2**	ℓ **= 2.5**	ℓ **= 3.0**
19	24.210133	24.210133	24.210133
30	24.630058	24.630058	24.630058
85	24.888991	24.888991	24.888991

Table 4: Joint KL divergence results for $k = 19, 30, 85$ - anonymization for $\ell = 2, 2.5, 3$

Interestingly, although we increased k to achieve ℓ-diversity by suppressing attributes like **Country** and **Gender**, the joint KL divergence did not increase and in some cases slightly decreased. We believe this may be due to the fact that for $k = 19, 30, 85$, attributes like **Country** or **Gender** that were effectively suppressed by inserting a value such as "Country" or "Male/Female/Other". Suppressing these columns drops these attributes from our quasi-identifier set and from our calculation for KL Divergence as well. Thus the previous values obtained for KL divergence for $k = 2, 3, 4$ and not comparable with the KL values for $k = 19, 30, 85$. However, high k-values indicate that there is large utility loss between the original dataset and our $k = 19, 35, 85$ anonymized datasets.

Additionally, the values for KL Divergence do not increase significantly between the different values of $k = 19, 30, 85$ despite stark difference in the size of equivalence classes. We believe that this may be due to the fact that so much

utility has been lost already that further generalization does not greatly impact our utility metric.

Discussion of Results

In our initial anonymization, we tested three values of k: $k = 2, 3$, and 4, using **Age Range, Gender, and Country** as quasi-identifiers (QIDs). State was excluded because its inclusion led to nearly all unique tuples in combination with other attributes, which would have required extreme generalization and drastically reduced utility.

To incorporate ℓ-diversity for the sensitive attribute **work_interfere**, we performed additional generalization on equivalence classes. This resulted in new, higher values of k (19, 30, and 85) to satisfy ℓ-diversity at $\ell = 2, 2.5,$ and 3. The need to increase k arises because ensuring ℓ-diversity requires each equivalence class to contain Entropy ℓ well-represented sensitive values, which was achieved by grouping more records together.

Overall, this project illustrates the trade-off between privacy and utility. Increasing k to satisfy ℓ-diversity provides stronger protection against inferential identity disclosure and sensitive value disclosure. However, it transforms our dataset into one that lacks granularity, which results in a large utility loss measured by KL Divergence and hinders the ability to gain meaningful insights from this data.